

ASSUMPTION UNIVERSITY
Vincent Mary School of Engineering, Science and Technology
CSX3007 QUIZII (70 points) Semester 2/2024 KEY

Instructions: A non-programmable calculator is allowed. It is a closed-book exam.

1. (6 points) Assume that a **3.5GHz** unpipelined processor executes a benchmark program with the following instruction mix and clock cycle count shown in the following Table:

Instruction	Instruction Count	Clock cycle count
Integer arithmetic	4500	3
Data transfer	3500	2
Floating point	2400	4
Control transfer	1500	5

Effective CPI = 3.16, MIPS rate = 1108, and Execution time (CPU time) = 10.743 x 10⁻⁶ sec.

2. (1 point) Processor **A** performs a given in **3** seconds while processor **B** requires **5** seconds for the same task. By what percentage is **A** faster than **B**? **40%**
3. (3 points) Consider **four jobs** being run on a corporate computer. We express the observations in a rate measure of jobs per hour, and these data are 0.5, 0.45, 0.53, and 0.43. What is the *central tendency* of these measurements in jobs per hour? **4/(1/0.5 + 1/0.45 + 1/0.53 + 1/0.43) = 0.476**
4. (4 points) Assume that two computers (A and B) execute four loops of a scientific program. The details of their clock cycles for each loop are shown in the **Table** below. For a particular benchmark, loop1 is executed 30 times, loop2 is executed 40 times, loop3 is executed 50 times, and loop4 is executed 60 times. What is the mean speedup of the computers (from A to B)?

Weighted Geometric mean = 1.5796

Loop	Comp A	Comp B	Speed	Weight
1	40	30	1.33	0.17
2	50	20	2.5	0.22
3	30	25	1.2	0.28
4	24	15	1.6	0.33

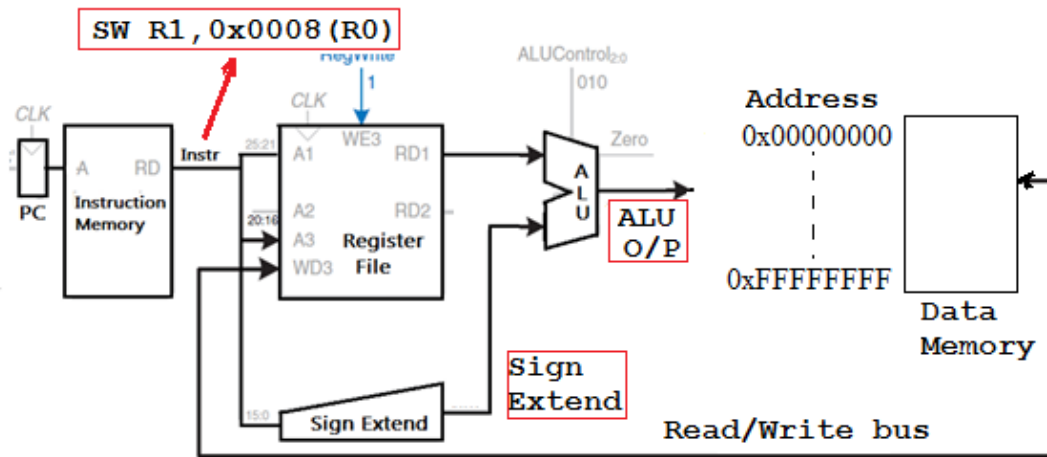
5. (4 points) A benchmark test on two sever systems, P4600 (an 8-core system) and H6600 (a 4-core system), has produced the following results: the P4600 shows 40% overhead during the test, whereas the H6600 has demonstrated best performance during 70% of its test time. So, which system is more efficient for running your task? **Speedup = 1/((1-a) + a/N), P4600: a = 0.6, N = 8, speedup = 2.11 H6600: a = 0.7, N = 4, speedup = 2.11. Both systems are efficient.**
6. (2 points) What is an **instruction set**? Why is it important? **Find out from the lecture slide**
7. (3 points) Briefly describe the **four state elements** of the MIPS system that support its Datapath. **Find out from the lecture slide**
8. (2 points) How does a CPU decode instructions as part of its execution? **After the compiler-generated instruction (macro instruction) is fetched from the Instruction memory, it is stored in the instruction register for decoding – comparing the macro instruction with the micro-instructions (instruction set) of the CPU, which is a part of the control unit. This macro and micro-instruction comparison would identify the user instruction, resulting in the generation of necessary signals to execute the instruction by the ALU.**
9. (6 points) Answer the following questions based on the single-cycle data path of the MIPS system shown in the **Figure** below. Assume that the registers **R0** and **R1** have initialized with **0x00000007** and **0xC123FAE7**: a). [1 point] Show the **O/P of the Sign Extend unit** (in hexadecimal) **0x00000008**
b). [1 point] Show the **O/P of the ALU** (in hexadecimal) **0x0000000F** c). [4 points] Show the result of the SW operation in Data Memory (DM).

0x0000000F: 0xC1

0x00000010: 0x23

0x00000011: 0xFA

0x00000012 : 0xE7



10. (4 points) Consider the following MIPS assembly code (where R0, R1, and R2 are 32-bit general purpose registers, and R0 has initialized with a value of **0x00000001**):

- i) **ADDI R0, R0, 0x0001**
- ii) **LW R1, 0x0003(R0)**
- iii) **SLL R0, R0, 0x0002**
- iv) **SW R2, 0x0002(R0)**

Based on the code sequence, answer the following: **(a).** (1point) Show the value (hexadecimal) of the register **R0** after the executions of the instructions (i) and (iii) **0x00000008** **(b).** (3 points) Show the **physical memory addresses** (hexadecimal) of the instructions (ii) **0x00000005** and (iv) **0x0000000A**

11. (2 points) A high-level language compiler running on a MIPS system has the following source code. Show its complete assembly code (where **g** and **h** are integer variables).

```

if (g > h)
    g = g + h;
else
    g = g - h;
LW R1, address g // address of g is not available
LW R2, address h // address of h is not available
BNG R1, R2, SUBTRACT // BNG means if R1 !> R2 ,
ADD R1, R1, R2
J EXIT
SUBTRACT: SUB R1, R1, R2
EXIT:
-----OR-----
LW R1, address g
LW R2, address h
BG R1, R2, ADDITION // BG means R1 > R2 ,
SUB R1, R1, R2
J EXIT
ADDITION: ADD R1, R1, R2
EXIT:
    
```

12. (3 points) Based on the given instruction sequence, determine the **values** (in hexadecimal) of the registers **R1**, **R2**, and **R3** after the instruction execution (assume that the register **R0** is initialized with **0x00000001**).

1. **ADDI R1, R0, 0x0005**
2. **SLL R1, R1, 1**
3. **ADDI R2, R0, 0x000A**
4. **BEQ R1, R2, NEXT**
5. **SUBI R1, R0, 8**
6. **ADD R3, R2, R0**
7. **NEXT: ADDI R3, R2, 0x0002**

R1 = 0xFFFFFFFF9

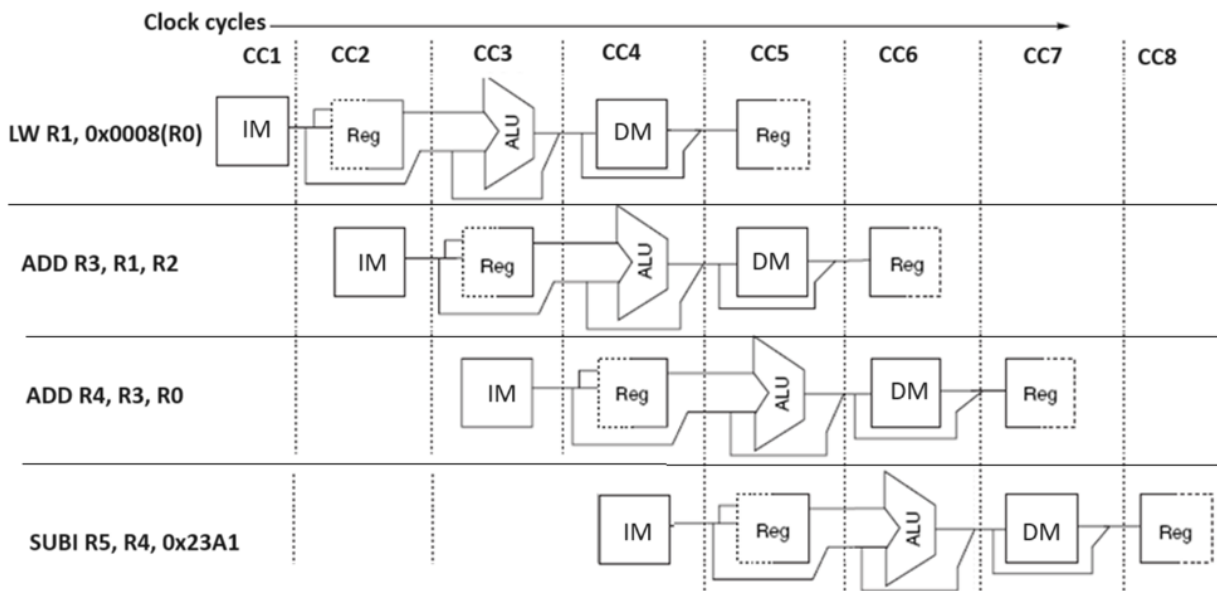
R2 = 0x0000000B

R3 = 0x0000000D

13. (4 points) What is **pipelining**? Describe the functions of various pipeline stages of the MIPS system.

Find this answer from the lecture slide

14. (2 points) Consider an *S-stage* pipeline system executing a program with *K* instructions. Show that the **speedup** of the pipeline is equivalent to the *K* when *S* reaches infinity. **Speedup = SK/(S+K-1), After dividing both numerator and denominator with S: K/(1 + (K-1)/S), if S → ∞, Speedup = K**
15. (4 points) Consider an **unpipelined** processor. It has a **2 ns** clock cycle and uses **four** cycles for ALU operations, **five** for branches, and **six** for memory operations. Assume that the relative frequencies of these operations are **50%, 15%, and 35%**, respectively. **Pipelining** the processor adds **0.4 ns** of overhead to the clock (ignore other pipeline latencies and hazards). Find out how much speedup the instruction execution rate will gain from the pipeline.
Avg.Inst.Exe.time(un-pipeline) = 9.7ns, Avg.Inst.Exe.time(pipeline) = 2.4ns.
Speedup = 9.7ns/2.4ns = 4.042 times
16. (2 points) Assume that the MIPS pipelined system has imposed a stall of **0.78** clock cycles per instruction for a particular user process. What is the **CPI** of the system? **1.78**
17. (3 points) What is a *structural hazard*? Consider the **pipeline data path** of the MIPS system shown in the **Figure** below. Is there any structural hazard in the system? If yes, describe the possible ways (if any) to eliminate it. **Resource conflicts in a pipeline execution are called structural hazards. There is no structural hazard with the pipeline operation given.**



18. (4 points) Consider the given MIPS instruction sequence:

		Clock Cycles ----->										
Instructions		1	2	3	4	5	6	7	8	9	10	11
Data hazard by R2	1. SUB(R2) R1, 0x98AF	IF	ID	EX	MEM	WB						
Data hazard by R4	2. AND(R4) R5, R2		IF	ID	EX	MEM	WB					
Data hazard by R3	3. ADD(R3) R4, R7			IF	ID	EX	MEM	WB				
Data hazard by R8	4. LW(R8) 0x3420(R3)				IF	ID	EX	MEM	WB			
	5. OR(R9) R8, R6					IF	ID	STALL	EX	MEM	WB	
Data hazard by R9	6. SW(R9) 0x4690(R8)						IF	STALL	ID	EX	MEM	WB

- Using the MIPS’s pipeline sequence diagram**, identify the **Read After Write (RAW)** data hazards among the instructions (if any) by indicating the **hazard-causing registers with their instruction numbers**. If any RAW data hazard is detected among instructions, show the various solution steps taken by the MIPS system (where R1, R2, R3, R4, R5, R6, R7, and R8 are 32-bit general purpose registers).
19. (3 points) Below is a section of the MIPS’s **instruction memory (IM)**. Based on that: (a). (1 point) What is the **Program Counter (PC)** value if the **IF stage** selects the instruction **0x5B4C6732**? **0x0000000C** b). (1 point) Show the value of the PC when the previous instruction was fetched **0x00000008** c). (1 point). Suppose the value of the PC is **0x0000000D**. Show the fetched instruction **0x4C6732AF**

Memory Address	Memory Word
0x00000008	0x23
0x00000009	0x5A
0x0000000A	0x1E
0x0000000B	0xA0
0x0000000C	0x5B
0x0000000D	0x4C
0x0000000E	0x67
0x0000000F	0x32
0x00000010	0xAF

20. (2 points) Why does the MIPS system use a **5-bit address** to access its register file? **There are 32 registers in the MIPS CPU.**
21. (6 points) Consider the following **high-level language code** running on a MIPS system. Show its MIPS *assembly code translation*, including *complete memory operations* with the **data memory (DM)** for the variable allocation. The size of the integer value in each variable is **32-bit**. The **32-bit memory address** of the first byte of the variable, *a*, is **0x00000008**. The variables *b* and *c* use the consecutive byte addresses for their values in DM memory. Use the 32-bit registers **R1**, **R2**, and **R3** for memory operations, and register **R0** as the base register and initialize it with a value of **0x00000004**).

```
int a = 10;
int b = 5;
int c = 16;
```

Data Memory

32-bit Address	Byte word
0x00000008	0X00
0x00000009	0X00
0x0000000A	0X00
0X0000000B	0X0A
0X0000000C	0X00
0X0000000D	0X00
0X0000000E	0X00
0X0000000F	0X05
0X00000010	0X00
0X00000011	0X00
0X00000011	0X00
0X00000012	0X10

int a = 10;
LW R1, 0x0004(R0)
R1 = 0x0000000A

int b = 5;
LW R2, 0x0008(R0)
R2 = 0x00000005

int c = 16;
LW R3, 0x000C(R0)
R3 = 0x00000010

===== End =====